THABANG MOLISE

Backend-Focused Software Engineering Student

LTMolise@outlook.com | +27 62-879-6640 | Pretoria, South Africa
lethasky.co.za

PROFILE

Backend-focused Software Engineering student in his final year at Belgium Campus iTversity, with hands-on experience building REST APIs, authentication systems, and relational/non-relational data layers across several team and personal projects. Comfortable working across the stack (Node.js, Express, PostgreSQL, Prisma, MongoDB, React), with a growing specialisation in computer vision and edge AI through robotics coursework. Currently leads the campus Robotics Club.

TECHNICAL SKILLS

Backend & APIs

- Node.js & Express
- JWT Authentication
- REST API Design
- PostgreSQL & Prisma
- MongoDB & Mongoose

Frontend & Tools

- React
- Git & GitHub
- Postman
- Object-Oriented Programming
- Agile / Kanban

AI & Systems

- Computer Vision (YOLOv8)
- Edge AI & Model Evaluation
- Monocular Depth (MiDaS)
- IoT / Embedded (Arduino)
- Problem Solving

PROJECTS

Trading Bot — Personal Project

Python | LightGBM, MetaTrader 5

- Built an AI-powered swing-trading bot for EURUSD/GBPUSD combining a multi-timeframe strategy (H4 directional bias, M15 entries) with LightGBM gradient-boosting models and market regime detection.
- Implemented a deterministic risk-management engine capping trade- and portfolio-level exposure, a high-impact news filter, and a quarterly model-refresh cycle; trades logged locally via SQLite.
- Iterated on the reward function to balance the Sortino ratio against reward-hacking behaviour in the decision engine.

Trade Tracker — Personal Project

React, Vite, Tailwind (PWA) | Node.js, Express, Prisma, Neon (PostgreSQL)

- Built a mobile-first trade-tracking web app, developed as both a portfolio piece and a working tool for personal trading.
- Implemented full trade CRUD functionality with JWT-based authentication and most core screens complete.
- Currently building aggregate API endpoints for dashboard and calendar summaries, ahead of wiring the frontend to the live backend, adding a Socket.IO real-time feed, and deploying to Vercel and Railway.

Classical Tracker (Vista Etude) — Personal Project

Flutter, Java, Stitch

- Designed and built a native single-user app for classical piano practice: local sheet music storage with a dedicated viewer and auto-scroll, practice session recording, and reference-recording storage.
- Architected the backend with a full Prisma schema, REST API endpoints, and file upload/streaming infrastructure.
- Kept the app deliberately account-free and local-first, with optional cloud backup, to match its single-user scope.

MJ Tribute Booking System — School Project (WPR381)

Node.js, Express, EJS, MongoDB, bcrypt

- Built the authentication and security foundation for a concert booking system, including session management with express-session and password hashing with bcrypt.

EDUCATION

Bachelor of Computing (Software Engineering) — In Progress *Belgium Campus iTversity, 2024 – present*

Final-year student specialising in Software Engineering.

Secondary Education *St. Andrew's School, Welkom — Matriculated 2023*

LEADERSHIP & ACHIEVEMENTS

- Head of Robotics Club, Belgium Campus iTversity (2025 – present).

LANGUAGES

English (Proficient) | Sesotho (Native) | Afrikaans (Intermediate) | Japanese (Elementary)

REFERENCES

Simba Zengeni — Projects Coordinator & Lecturer, Belgium Campus iTversity | +27 67-100-7329 | zengeni.s@belgiumcampus.ac.za

Thabang Lebopa — Student Advisor, Belgium Campus iTversity | +27 72-251-6330 | thabang.lebopa@gmail.com